

PRANAV R

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PROFESSIONAL SUMMARY

Systems Engineer with hands-on experience building infrastructure observability pipelines, real-time telemetry systems, and alerting workflows in a production Go environment. Proven ability to monitor distributed system health, identify performance bottlenecks, support incident response, and maintain dashboards that surface actionable insights. Strong Linux foundation with backend development experience. Eager to grow into an infrastructure engineering role with genuine curiosity for systems reliability and operational automation.

PROFESSIONAL EXPERIENCE

System Engineer | Karios

May 2025 – Present

Infrastructure Observability | Telemetry Pipelines | Prometheus · InfluxDB | Go | Linux

- Built a real-time telemetry ingestion pipeline in Go, parsing live cellular metrics (RSRP, RSRQ, SINR) from Sierra Wireless 4G modems via mmcli and scraping WAN interface metrics from node_exporter textfiles; integrated into the karios-edge daemon.
- Monitored distributed system performance across multi-WAN SD-WAN deployments, proactively identifying degradation events and triaging root causes using log correlation and metrics analysis.
- Implemented automated alerting and event-driven routing logic — OSPF cost injection via FRRouting triggered by threshold breaches — reducing recovery time for WAN link failures.
- Built backend API services in Go including RBAC middleware and authentication flow integration, contributing to reliable, observable backend infrastructure.
- Performed Linux systems diagnostics — namespace management, NIC/sysfs-level inspection, iproute2 routing validation, WireGuard tunnel debugging — maintaining operational stability.

Technology Support Intern | ICITY Technologies

Mar 2025 – May 2025

Raspberry Pi 4 | Linux | Node.js | MagicMirror²

- Implemented and customized the MagicMirror open-source platform on Raspberry Pi 4, extending it with modules for real-time weather, calendar, system status, voice commands, and Spotify integration.
- Gained practical exposure to Linux-based single-board computing, Node.js module development, and voice recognition integration in a production-adjacent environment.

TECHNICAL SKILLS

Monitoring & Observability: Prometheus, InfluxDB, Node_exporter, Alerts, Metrics aggregation

Linux & Systems: Linux administration, iproute2, Namespace management, sysfs/NIC diagnostics, Bash scripting

Programming: Go, Python, C, C++

Networking: OSPF, WireGuard, FRRouting, MQTT, WiFi, BLE, Ethernet, Layer 2/3 concepts

Backend & APIs: Go REST services, RBAC middleware, Authentication flows, Structured logging

Embedded & IoT: Raspberry Pi, ESP32, ESP8266, ARM Cortex, Arduino

Tools & Workflow: Git & GitHub, Zoho Sprints, Agile/Scrum, Operational runbooks

EDUCATION

B.E. Electronics and Communication Engineering

November 2021 – May 2025

KGISL Institute of Technology — CGPA: 7.6 / 10.0

PROJECTS

8-bit Computer – Embedded Architecture Design

Jul 2023 – Mar 2024

Arduino IDE, C++, Logic ICs, EEPROM, ALU Components | Hardware Designer

- Designed a complete 8-bit processor from discrete components (flip-flops, counters, resistors) with ALU, memory interface, and control unit, programmed using a custom assembly-like instruction set.

Password Protected Lock System using ESP32

Jan 2025 – Apr 2025

ESP32, I2C, C++, MFCC | Independent Developer

- Built on-device speaker verification with MFCC feature extraction and PIN-based fallback — demonstrating multi-layer authentication and fail-safe access control logic.

Voice Control Home Automation & Sensor Monitoring

June 2023

ESP8266, MQTT, Mosquitto Broker | KGXperience IoT Internship

- Deployed a distributed IoT monitoring system using MQTT pub-sub — collecting sensor data in real time and triggering responses based on threshold events, analogous to infrastructure alerting.